

Some of the questions that were covered in class and office hours

Problem 4- Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the maximum likelihood estimator (MLE) for θ .

(b) Find the method of moments estimator (MOM) for θ .

(b)

$$E[X] = \int_{-\infty}^{\infty} x f_{\theta}(x) dx$$

$$E[X] = \frac{1}{\theta} \int_0^1 x^{\frac{1}{\theta}} dx$$

$$E[X] = \frac{1}{\theta+1} x^{\frac{\theta+1}{\theta}} \Big|_0^1 = \frac{1}{\theta+1}$$

By method of moments

$$\bar{X}_n = \frac{1}{\theta+1}$$

$$\theta \bar{X}_n + \bar{X}_n = 1$$

$$\tilde{\theta} = \frac{1 - \bar{X}_n}{\bar{X}_n}$$

(a)

$$\begin{aligned}L(x, \theta) &= \prod_{i=1}^n f_X(x_i|\theta) \\L(x, \theta) &= \prod_{i=1}^n \frac{1}{\theta} x_i^{\frac{1-\theta}{\theta}} \\ \ln(L(x, \theta)) &= \sum_{i=1}^n \left(\ln\left(\frac{1}{\theta}\right) + \ln\left(x_i^{\frac{1-\theta}{\theta}}\right) \right) \\ \ln(L(x, \theta)) &= \sum_{i=1}^n \left(\ln\left(\frac{1}{\theta}\right) + \frac{1-\theta}{\theta} \ln(x_i) \right) \\ \ln(L(x, \theta)) &= -n \ln(\theta) + \frac{1-\theta}{\theta} \sum_{i=1}^n \ln(x_i) \\ \ln(L(x, \theta)) &= -n \ln(\theta) + \left(\frac{1}{\theta} - 1\right) \sum_{i=1}^n \ln(x_i)\end{aligned}$$

Now we take derivative with respect to θ ,

$$\begin{aligned}\frac{d}{d\theta}(\ln(L(x, \theta))) &= \frac{-n}{\theta} - \frac{1}{\theta^2} \sum_{i=1}^n \ln(x_i) = 0 \\ \hat{\theta} &= -\frac{\sum_{i=1}^n \ln(x_i)}{n}\end{aligned}$$

Problem 3. Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} (\theta - 1)^2 \frac{\ln(x)}{x^{\theta}} & x > 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) In this part, assume $\theta > 1$. Find the maximum likelihood estimator (MLE) for θ .

(b) In this part, assume $\theta > 2$. Find the method of moments estimator (MOM) for θ .

(a)

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n (\theta - 1)^2 \frac{\ln x_i}{x_i^\theta} \\ \ln L(\theta) &= 2n \ln(\theta - 1) + \sum_{i=1}^n \ln(\ln x_i) - \theta \sum_{i=1}^n \ln(x_i) \\ \frac{d \ln L(\theta)}{d\theta} &= \frac{2n}{\theta - 1} - \sum_{i=1}^n \ln(x_i) = 0 \end{aligned}$$

By solving the above equation for θ we have

$$\hat{\theta} = 1 + \frac{2n}{\sum_{i=1}^n \ln(X_i)}$$

(b)

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_1^{\infty} x(\theta - 1)^2 \frac{\ln x}{x^\theta} dx = \frac{(\theta - 1)^2}{(\theta - 2)^2}$$

Now by method of moments

$$\begin{aligned} E[X] &= \bar{X} \\ \frac{(\theta - 1)^2}{(\theta - 2)^2} &= \bar{X} \end{aligned}$$

By solving the above equation for θ we obtain,

$$\tilde{\theta} = \frac{2\sqrt{\bar{X}} - 1}{\sqrt{\bar{X}} - 1}$$

To solve the integral $\int_1^{\infty} x(\theta - 1)^2 \frac{\ln x}{x^\theta} dx = (\theta - 1)^2 \int_1^{\infty} \frac{\ln x}{x^{\theta-1}} dx = (\theta - 1)^2 \int_1^{\infty} (\ln x) x^{1-\theta} dx$. We should use integration by parts.

$$\begin{aligned} u &= \ln x & du &= \frac{dx}{x} \\ dv &= x^{1-\theta} dx & v &= \frac{x^{2-\theta}}{2-\theta} \end{aligned}$$

Integration by parts tells $\int u dv = uv - \int v du$, so we have

$$\begin{aligned} (\theta - 1)^2 \int_1^{\infty} (\ln x) x^{1-\theta} dx &= (\theta - 1)^2 (\ln x) \frac{x^{2-\theta}}{2-\theta} \Big|_1^{\infty} - (\theta - 1)^2 \int_1^{\infty} \frac{x^{2-\theta}}{2-\theta} \frac{dx}{x} \\ &= \lim_{x \rightarrow \infty} \frac{(\ln x)}{x^{\theta-2}} \frac{(\theta - 1)^2}{2-\theta} - \frac{(\theta - 1)^2}{2-\theta} \int_1^{\infty} x^{1-\theta} dx \end{aligned}$$

We can use L'hopital rule to solve the limit,

$$(\theta - 1)^2 \int_1^\infty (\ln x) x^{1-\theta} dx = - \frac{(\theta - 1)^2}{(2 - \theta)^2} \lim_{x \rightarrow \infty} \frac{1}{x^{\theta-2}} - \frac{(\theta - 1)^2}{(2 - \theta)^2} x^{2-\theta} \Big|_1^\infty = 0 + \frac{(\theta - 1)^2}{(2 - \theta)^2} = \frac{(\theta - 1)^2}{(\theta - 2)^2}$$

Problem 12. During one day Sarah receives X legitimate emails and Y spam emails. Suppose X and Y are independent, $X \sim Pois(\lambda_x)$ and $Y \sim Pois(\lambda_y)$. Assuming one day Sarah receives n emails. Given this information what is the expected value of how many legitimate email she has? Show all the work.

X: legitimate emails, Y: spam emails

$$X \sim \text{Poisson}(\lambda_x), \quad Y \sim \text{Poisson}(\lambda_y)$$

$$T = X + Y$$

$$P(T = n) = \sum_{k=0}^n P(X = k)P(Y = n - k)$$

$$= \sum_{k=0}^n \frac{e^{-\lambda_x} \lambda_x^k}{k!} \cdot \frac{e^{-\lambda_y} \lambda_y^{n-k}}{(n-k)!}$$

$$= e^{-(\lambda_x + \lambda_y)} \sum_{k=0}^n \frac{\lambda_x^k \lambda_y^{n-k}}{k!(n-k)!} = e^{-(\lambda_x + \lambda_y)} \cdot \frac{(\lambda_x + \lambda_y)^n}{n!}$$

So,

$$T \sim \text{Poisson}(\lambda_x + \lambda_y)$$

Conditional distribution:

$$P(X = k | T = n) = \frac{P(X = k, Y = n - k)}{P(T = n)}$$

Since X and Y are independent, we can rewrite the right hand side as follows

$$P(X = k | T = n) = \frac{P(X = k)P(Y = n - k)}{P(T = n)} = \frac{\frac{e^{-\lambda_x} \lambda_x^k}{k!} \cdot \frac{e^{-\lambda_y} \lambda_y^{n-k}}{(n-k)!}}{\frac{e^{-(\lambda_x + \lambda_y)} (\lambda_x + \lambda_y)^n}{n!}}$$

$$= \frac{n!}{k!(n-k)!} \cdot \left(\frac{\lambda_x}{\lambda_x + \lambda_y} \right)^k \left(\frac{\lambda_y}{\lambda_x + \lambda_y} \right)^{n-k} = \binom{n}{k} \cdot \left(\frac{\lambda_x}{\lambda_x + \lambda_y} \right)^k \left(1 - \frac{\lambda_x}{\lambda_x + \lambda_y} \right)^{n-k}$$

So:

$$X \mid (T = n) \sim \text{Binomial} \left(n, \frac{\lambda_x}{\lambda_x + \lambda_y} \right)$$

$$\mathbb{E}[X \mid T = n] = n \cdot \frac{\lambda_x}{\lambda_x + \lambda_y}$$

Problem 7. Let X and Y be two independent random variables with the CDFs F_X and F_Y . Define

$$\begin{aligned} Z &= \max(X, 2Y), \\ W &= \min(X, Y) \end{aligned}$$

Find the CDFs of Z and W .

To calculate CDF of $Z = \max(X, 2Y)$, step one we write down definition of cdf.

$$F_Z(z) = P(\max(X, 2Y) \leq z) = P(X \leq z \text{ and } 2Y \leq z)$$

$$= P(X \leq z) \cdot P(2Y \leq z) \quad (\text{independence})$$

$$= F_X(z) \cdot P\left(Y \leq \frac{z}{2}\right) = F_X(z) \cdot F_Y\left(\frac{z}{2}\right)$$

So,

$$F_Z(z) = F_X(z) \cdot F_Y\left(\frac{z}{2}\right)$$

To calculating CDF of $W = \min(X, Y)$, by using definition of CDF, we write

$$F_W(w) = P(\min(X, Y) \leq w) = 1 - P(X > w \text{ and } Y > w)$$

$$F_W(w) = 1 - P(X > w) \cdot P(Y > w) = 1 - (1 - F_X(w))(1 - F_Y(w)) = 1 - (1 - F_X(w) - F_Y(w) + F_X(w)F_Y(w))$$

$$F_W(w) = F_X(w) + F_Y(w) - F_X(w)F_Y(w)$$